

Question 3

Class Result Processing

Given Data

- Number of students (N)
- 5 Subjects for each student
- Each marks is out of 100

Required Result

- Average of each student
- Result (Distinction/Pass/fail)
- Subject Deficiency status.

Process Required Alternative Solution

- Input N
Use array to store marks 5 subjects marks.
- Process N students using an outer loop
Use of for loop instead of while.
- for each student, process 5 subjects using an inner loop
- Add the 5 subject marks.
- calculate Average
- check whether any subject mark is < 33
if yes set result "fail"
- otherwise check if average ≥ 80 result is Distinction, otherwise if average ≥ 60 result is Pass
- otherwise result is fail

Date:

class result processing

IPO Chart

Input

Process

Output

- | Input | Process | Output |
|-------------------------------|--|---|
| - Number of students (N) | Repeat for N students | Student average marks. |
| - 5 subjects for each student | initialize obtained = 0 and deficiency = 0 | Student result |
| - | Read 5 marks from inner loop | "Fail - Subject deficiency" if deficiency = 1 |
| - | Add each marks to obtained | |
| - | check if any mark < 33 | |
| - | Calculate average | |
| - | Classify based on average | |
| - | Override result if any subject < 33 | |

Pseudocode

- Begin
- Declare $n, i, j = 0$, subjects=5, marks, obtained=0, deficiency=0, as integer and average as float.
- Input n
- while ($n > i$)
 - $i \leftarrow i + 1$
 - while (subjects $> j$)
 - $j \leftarrow j + 1$
 - input marks
 - Obtained $\leftarrow$ obtained + marks
 - if (marks < 33)
 - deficiency = 1;
 - end if
 - end while
 - average $\leftarrow$ obtained / 5.0
 - Display average
 - if (deficiency ≥ 1)
 - Display "fail - Subject Deficiency".
 - elseif (average ≥ 80)
 - Display "Distiction"
 - elseif (average ≥ 60)
 - Display "Pass"
 - else
 - Display "Fail"
 - end if

end while

end

Algorithm

1. Start
2. Take input number of students (N) from the user.
3. Set $i=0$, $j=0$, deficiency $=0$, subjects $=5$
4. Repeat the following steps while $(N > i)$
 1. Set obtained $=0$
 2. increment i by 1
 3. Repeat the following steps while $(\text{subjects} > j)$
 1. increment j by 1
 2. Take input, marks from the user.
 3. Set obtained $= \text{obtained} + \text{marks}$.
 4. if marks < 33 ,
 - set deficiency $= 1$
 4. Calculate average by $\text{average} = \text{obtained} / 5$
 5. Display average on the screen
 6. if deficiency $= 1$
 - Display "fail-Subject Deficiency" on the screen
 7. else if average ≥ 80
 - Display "Distinction" on the screen
 8. else if average ≥ 60
 - Display "Pass" on the screen
 9. else
 - Display "Fail" on the screen.
5. Stop

Flowchart

3. Class result processing.



